POTENTIAL OUTCOMES FRAMEWORK

$Y_i(1)$ = outcome if treated, $Y_i(0)$ = outcome if control. These are **fixed numbers**, not random variables. Only one is ever observed per person.
$\tau_i = Y_i(1) - Y_i(0)$ (individual treatment effect – NEVER observable)
$Y_i^{\text{obs}} = Z_i Y_i(1) + (1 - Z_i) Y_i(0)$ (switching equation)

$$\tau = \frac{1}{N}\sum_{i=1}^{N}[Y_i(1) - Y_i(0)] \quad \textbf{(ATE – a fixed number)}$$

$\hat{\tau} = \bar{Y}_T - \bar{Y}_C$ (difference-in-means estimator)
Under CRD: $\mathbb{E}[\hat{\tau}] = \tau$ (unbiased – follows from randomization alone)
Why unbiased: each person is equally likely to be in the treated group, so $\bar{Y}_T$ is unbiased for $\frac{1}{N}\sum Y_i(1)$ and $\bar{Y}_C$ for $\frac{1}{N}\sum Y_i(0)$. Subtract $\Rightarrow$ done. No modeling needed.
ATT $= \mathbb{E}[Y(1) - Y(0) \mid Z = 1]$ **ATC** $= \mathbb{E}[Y(1) - Y(0) \mid Z = 0]$
In an RCT: ATE = ATT = ATC (groups are comparable by design).
The big idea: potential outcomes = fixed (science). Z = random (assignment). ALL probability comes from randomization only. This is "design-based" inference – no model for outcomes needed.

NEYMAN VARIANCE

$S_z^2 = \frac{1}{N-1}\sum(Y_i(z) - \bar{Y}(z))^2$ for $z \in \{0, 1\}$
$S_\tau^2 = \frac{1}{N-1}\sum(\tau_i - \tau)^2$ (how much the effect varies across people)

$$\textbf{Exact:}\ \mathrm{Var}(\hat{\tau}) = \frac{S_1^2}{n_1} + \frac{S_0^2}{n_0} - \frac{S_\tau^2}{N}$$

Reading it: more spread in POs (S_1^2, S_0^2) = noisier. Bigger groups = less noise. Surprisingly, more heterogeneity (S_τ^2) = LESS variance (high-τ people in treated push $\bar{Y}_T$ up and their absence pushes $\bar{Y}_C$ down – the difference doesn't move as much).
$S_{10} = \frac{1}{N-1}\sum(Y_i(1) - \bar{Y}(1))(Y_i(0) - \bar{Y}(0))$ – requires both POs $\Rightarrow$ **unidentifiable**. This is the fundamental unidentifiable quantity: do people who do well treated also do well untreated?
Identity: $S_\tau^2 = S_1^2 + S_0^2 - 2S_{10}$

$$\textbf{Conservative:}\ \widehat{\mathrm{Var}}(\hat{\tau}) = \frac{s_1^2}{n_1} + \frac{s_0^2}{n_0}$$

Drops $-S_\tau^2/N$. Since $S_\tau^2 \geq 0$: estimate too big $\Rightarrow$ CIs too wide $\Rightarrow$ coverage $\geq 95\%$. **Always valid, never too narrow.**
Exact when: constant effects ($\tau_i = \tau\ \forall i$, so $S_\tau^2 = 0$). Proof: $Y_i(1) = Y_i(0) + \tau \Rightarrow S_1^2 = S_0^2$, $S_{10} = S_0^2 \Rightarrow S_\tau^2 = 0$.

95% CONFIDENCE INTERVAL

$$\hat{\tau} \pm 1.96\sqrt{s_1^2/n_1 + s_0^2/n_0}$$

The 4-step logic chain: (1) unbiasedness centers $\hat{\tau}$ at τ, (2) FP-CLT $\Rightarrow$ approx Normal (from randomization, NOT assuming outcomes are normal), (3) ± 1.96 SD covers 95% of Normal, (4) SE is conservative (dropped $S_\tau^2/N \geq 0$) so coverage $\geq 95\%$. Works for ANY population, ANY outcome distribution.

FP-CLT vs i.i.d. CLT

FP-CLT (this class)	i.i.d. CLT (intro stats)
Only Z_i is random	Everything is random
Outcomes are FIXED numbers	Outcomes = random draws
Finite population of N	Infinite superpopulation
Variance from assignment	Variance from sampling
No normality assumption	Needs normality or large n
Does NOT pool variances	Often pools variances

Neyman's approach uses the top column. Classical t-test uses the bottom. Similar answers when $n_1 \approx n_0$ and variances similar, but justification is totally different.

FISHER vs NEYMAN

Fisher	Neyman
Test if $\tau = 0$ (p-value)	Estimate τ (point est + CI)
Sharp null (specifies all τ_i)	No assumption on τ_i
Exact (enumerate assignments)	Asymptotic (CLT-based)
Struggles w/ heterogeneity	Targets average naturally

Complementary: Fisher = "is there any effect?" Neyman = "how big, and how uncertain?"

FWL THEOREM

$\tilde{Z}_i = Z_i - \hat{\mathbb{E}}[Z \mid X_i]$ (residualize Z on X)

$$\hat{\tau}_{\text{FWL}} = \frac{\sum \tilde{Z}_i Y_i}{\sum \tilde{Z}_i^2} = \text{coeff on } Z \text{ in } Y \sim Z + X$$

Weights: $w_i = \tilde{Z}_i / \sum \tilde{Z}_j^2$
Balancing: $\sum w_i X_i = 0$ (regression *creates* balance on X), $\sum_{Z_i=1} w_i = 1$
Misspecification bias: Bias $= \sum w_i\, g(X_i)$
Intuition: regression balances X linearly. But if the true model has X^2, $\log X$, or unobserved confounders, $\sum w_i g(X_i) \neq 0$ and you get bias. Example: $\bar{X}_T = \bar{X}_C$ but treated have high $X^2 \Rightarrow$ biased.
Extrapolation: groups in different X regions $\Rightarrow$ regression guesses across the gap. No data in between means pure model-dependent fiction.
When mispec doesn't matter: if $\mathbb{E}[Z \mid X] = \mathbb{E}[Z]$ (e.g., RCT), then $\tilde{Z} \perp g(X) \Rightarrow$ bias = 0.
Regression in experiments: unbiased even if model wrong (randomization $\Rightarrow \tilde{Z} \perp g(X)$). Reduces variance like noise-cancelling headphones. Does NOT change estimand. Refines a good design, doesn't rescue a bad one.

FISHER'S FRAMEWORK

Sharp null: $H_0 : Y_i(1) = Y_i(0)$ for ALL i. This *fills in every missing outcome*: under H_0, $Y_i(1) = Y_i(0) = Y_i^{\text{obs}}$ for everyone. That is what makes the test possible.
Weak null: $H_0 : \tau = 0$. Can't fill in individual POs $\Rightarrow$ can't run randomization test.
Sharp false + weak true: possible! E.g., $\tau_i = +1, +1, -1, -1$. Average = 0 (weak true) but every individual has $Y_i(1) \neq Y_i(0)$ (sharp false).
The randomization test (know these steps cold): (1) State H_0: sharp null. (2) Under H_0, all POs known $= Y^{\text{obs}}$. (3) Choose test stat T (e.g., diff-in-means). (4) Compute T_{obs}. (5) Enumerate all $\binom{N}{n_1}$ assignments, compute T for each. (6) p-value = fraction with $|T| \geq |T_{\text{obs}}|$.
Min p-value $= 1/\binom{N}{n_1}$. Ex: $N = 6, n_1 = 3 \Rightarrow 1/20 = 0.05$. With tiny N, some significance levels are literally impossible.
Exact, no distributional assumptions. Doubling one-sided p NOT always valid (randomization distribution can be asymmetric).
Test stats: diff-in-means (powerful, outlier-sensitive) vs Wilcoxon/ranks (robust). Different stats can give different p-values under the same sharp null because they weight the data differently.
Randomization test $\neq$ permutation test: same calculation, different justification (design-based vs model-based).
CI by inversion: adjust $Y^{\text{adj}} = Y^{\text{obs}} - Z\tau_0$, test each τ_0: $\mathcal{C} = \{\tau_0 : p(\tau_0) > \alpha\}$. *Idea:* if true $\tau = \tau_0$, subtracting it out should leave no treatment effect. Keep all values you can't reject. Exact, any sample size, no normality.

RANDOMIZATION DESIGNS

CRD: $\Pr(\mathbf{Z} = \mathbf{z}) = 1/\binom{N}{n_1}$. Each: $\Pr(Z_i = 1) = n_1/N$. Assignments NOT independent (fixing n_1 creates dependence).
Blocked: randomize within strata. $\hat{\tau}_{BL} = \sum_k \frac{n_k}{N}(\bar{Y}_{T,k} - \bar{Y}_{C,k})$. Removes between-block variation $\Rightarrow$ lower variance. $K = 1$ block = CRD.
Paired: strongest blocking. Match pairs, flip coin within each.
Bernoulli: $Z_i \sim \mathrm{Bern}(p)$ iid. Group sizes random. Can get very imbalanced.
Cluster: randomize whole clusters. Effective N = # clusters, not individuals.
Deterministic assignment bias: $Z_i = \mathbf{1}\{X_i > c\}$

$$\mathbb{E}[\bar{Y}_T - \bar{Y}_C] = \tau + \beta(\mathbb{E}[X \mid X > c] - \mathbb{E}[X \mid X \leq c])$$

Biased by construction. This is exactly what randomization prevents.

THREE ASSUMPTIONS (OBS STUDIES)

1. SUTVA: (a) no interference – my outcome depends only on MY treatment (b) no hidden versions of treatment. *When violated:* vaccines (herd immunity), social networks, classrooms.
2. Ignorability: $(Y(1), Y(0)) \perp\!\!\!\perp Z \mid X$ – conditional on X, treatment is as-good-as-random. **Untestable from data!** Must argue from subject-matter knowledge. This is the assumption you can never verify.
3. Overlap: $0 < \Pr(Z = 1 \mid X = x) < 1\ \forall x$. *If all young men are treated and none are control, you have no comparison for young men.*
Strong ignorability = #2 + #3 together.

SELECTION BIAS DECOMPOSITION

$$\mathbb{E}[Y \mid Z=1] - \mathbb{E}[Y \mid Z=0] = \underbrace{\text{ATT}}_{\text{causal}} + \underbrace{\mathbb{E}[Y(0) \mid Z=1] - \mathbb{E}[Y(0) \mid Z=0]}_{\text{selection bias}}$$

Derive it: SUTVA $\Rightarrow$ LHS $= \mathbb{E}[Y(1) \mid Z=1] - \mathbb{E}[Y(0) \mid Z=0]$. Add/subtract $\mathbb{E}[Y(0) \mid Z=1]$ and regroup. Selection bias = "would treated have had different outcomes even WITHOUT treatment?" *If exercisers are healthier at baseline, their $Y(0)$ is already higher $\Rightarrow$ naive estimate inflated.*

IDENTIFICATION PROOF (know this chain!)

$\mathbb{E}[Y(1)] = \mathbb{E}[\mathbb{E}[Y(1) \mid X]]$ (tower property)
$= \mathbb{E}[\mathbb{E}[Y(1) \mid Z=1, X]]$ (ignorability lets us add $Z=1$)
$= \mathbb{E}[\mathbb{E}[Y \mid Z=1, X]]$ (consistency + overlap)

Same for $Y(0)$. Subtract: $\boxed{\tau = \mathbb{E}[\mathbb{E}[Y \mid Z=1, X] - \mathbb{E}[Y \mid Z=0, X]]}$

This is the key result: it turns unobservable causal quantities into observable conditional means.

ATT: average inner part over $X \mid Z=1$. ATC: over $X \mid Z=0$. Works with $e(X)$ replacing X.

PROPENSITY SCORE

$e(X) = \Pr(Z=1 \mid X)$ (estimated via logistic regression)

Balancing property: $(Y(1), Y(0)) \perp\!\!\!\perp Z \mid e(X)$. *Collapse all covariates into one number and match on that instead.*

Proof: Fix $e(X)=p$. Everyone has $\Pr(Z=1)=p$, so treated are a random subset within this stratum. The p's cancel: $\Pr(X=x \mid Z=1, e=p) = \Pr(X=x \mid e=p)$.

PS coefficients = what predicts SELECTION into treatment, NOT causal effects. *A positive coeff on "married" means married people are more likely to get treated, not that marriage causes the outcome.* Goal = balance, not prediction. Don't include instruments (affect Z not Y – adds variance, no bias reduction).

BALANCE DIAGNOSTICS

$$\text{SMD} = \frac{\bar{X}_T - \bar{X}_C}{\sqrt{(s_T^2 + s_C^2)/2}}$$

< 0.1 good, ≈ 0.2 marginal, > 0.25 bad. Variance ratio $= s_T^2/s_C^2$, want ≈ 1.

Use SMD not t-tests: t-tests are sample-size dependent (huge N makes trivial differences "significant"; small N misses big ones). SMD measures actual substantive imbalance. Love plot: SMDs before vs after for each covariate – you want all dots moving toward zero.

DESIGN vs ANALYSIS (sacred boundary)

Design (no Y): estimate PS, match/weight, check balance (SMDs, Love plot), overlap (PS histograms). Iterate. **Analysis (use Y):** estimate $\hat{\tau}$, CI. *Never peek at outcomes during design.* This prevents fishing/p-hacking. Like pre-registration. Peeking at Y to choose matches = selection bias.

MATCHING

A design tool (NO outcomes). Creates balanced dataset $\Rightarrow$ simple analysis.

$$\hat{\tau} = \frac{1}{n_M} \sum_{Z_i=1} [Y_i - Y_{j(i)}] \quad j(i) = \text{matched control}$$

Intuition: if matched pair has similar X, background effects $g(X)$ cancel when you subtract, leaving mostly treatment effect.

Bias bound: $|\text{Bias}| \leq L(|\Delta_X^M| + 2s_D)$. L = steepness of g (nature, can't control). Δ_X^M = avg match gap, s_D = inconsistency of gaps (both under your control via better matching). *This is why we obsess over SMDs – they directly check the conditions that keep bias small.*

Mahalanobis: $d_M = \sqrt{(X_i - X_j)^\top \hat{\Sigma}^{-1} (X_i - X_j)}$ – rescales by spread + correlation. A 1-year age gap and a \$1 income gap are very different; Mahalanobis puts them on a fair scale. Breaks in high dimensions ("nearest" neighbor is still far) $\Rightarrow$ add PS caliper.

Composite: $d = (X_i - X_j)^2 + \lambda \cdot \mathbf{1}\{W_i \neq W_j\}$. Large λ = soft exact match on category. **Optimal:** minimizes TOTAL distance across all pairs (greedy NN doesn't).

1:k: more controls = lower variance, worse match quality. Tradeoff.

Exact match on categories can hurt continuous balance (forces bad matches just to stay within-group). **Near-fine balance:** softer – similar marginal distributions, allows cross-category matches. Better overall balance.

SUBCLASSIFICATION

$\hat{\tau}_{\text{strat}} = \sum_k \frac{N_k}{N} (\bar{Y}_{T,k} - \bar{Y}_{C,k})$. 5 PS strata $\approx$ 90% bias removed. Sparse strata ($n_1 = 1$) = high variance $\Rightarrow$ merge. If PS model wrong, stratification can make things WORSE.

IPW (INVERSE PROBABILITY WEIGHTING)

$$\hat{\tau}_{\text{IPW}} = \frac{1}{N} \sum \left[\frac{Z_i Y_i}{\hat{e}(X_i)} - \frac{(1-Z_i)Y_i}{1-\hat{e}(X_i)} \right]$$

Idea: a treated person with $e = 0.1$ "represents" many untreated people like them, so upweight. A treated person with $e = 0.9$ is typical of the treated population, low weight. Reweighting creates a pseudo-population where treatment is independent of X.

Why unbiased: $\mathbb{E}[ZY/e(X)] = \mathbb{E}[Y(1)]$. Tower + ignorability: $e(X)$ cancels perfectly.

Balances ANY $g(X)$: $\mathbb{E}[Zg(X)/e(X)] = \mathbb{E}[g(X)]$. Works for $X, X^2, \sin(X)$. *Unlike regression, IPW balances nonlinear functions too.* If empirical balance fails $\Rightarrow$ PS model is wrong.

$\mathbb{E}[w_i] = 2$: tower gives $\mathbb{E}[Z/e \mid X] = 1$, $\mathbb{E}[(1-Z)/(1-e) \mid X] = 1$. Sum = 2.

ESS $\approx (\sum w_i)^2 / \sum w_i^2$. Equal weights: N. Extreme: collapses. Ex: $N = 200$, half $e = 0.5$ (wt 2), half $e = 0.01$ (wt 100) $\Rightarrow$ ESS ≈ 15. *A few people dominate the whole estimate.*

Fixes: (1) **Trim** to $[0.1, 0.9]$ – changes estimand. (2) **Stabilized:** $w^s = Z\hat{p}/\hat{e} + (1-Z)(1-\hat{p})/(1-\hat{e})$. (3) **Hajek:** normalize weights per group. Tiny bias, much lower variance.

Matching can be stable when IPW is not: if most units have $e \approx 0.5$ (good matches) but a few have $e \approx 0.01$ (extreme weights), matching ignores the extreme units while IPW is destroyed by them.

AIPW (DOUBLY ROBUST)

$$\hat{\tau} = \frac{1}{N} \sum \left[\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) + \frac{Z_i(Y_i - \hat{\mu}_1)}{\hat{e}} - \frac{(1-Z_i)(Y_i - \hat{\mu}_0)}{1-\hat{e}} \right]$$

4 terms: $\hat{\mu}_1 - \hat{\mu}_0$ = regression guess. Remaining terms = IPW-weighted residuals (correction). If outcome model perfect $\Rightarrow$ residuals = 0, just regression. If PS model perfect $\Rightarrow$ correction rescues bad regression. **Doubly robust:** works if EITHER correct. Fails only if BOTH wrong. Best efficiency when both correct. Can still perform poorly in finite samples with extreme weights or bad fits.

Reg-based ATE: fit $\hat{\mu}_1$ on treated, $\hat{\mu}_0$ on controls, predict for everyone: $\hat{\tau} = \frac{1}{N} \sum [\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i)]$.

OVERLAP / ALL METHODS

PS near 0 or 1 $\Rightarrow$ ALL methods fail (not just IPW). Trim to $[0.1, 0.9]$ – changes estimand to overlap subpop. Check PS histograms: if treated and control barely touch, honest answer is "ATE cannot be reliably estimated."

ALL methods (matching, strat, IPW, AIPW, regression) need SUTVA + ignorability + overlap. **Unobserved confounding $\Rightarrow$ ALL wrong.** No amount of clever modeling fixes what you didn't measure. AIPW preferred: two chances to get it right.

R CODE / METHOD SUMMARY

PS: logistic reg of Z on covariates. Coefficients = selection, NOT causal.

Distance: Mahalanobis + PS with weights $c(w_1, w_2)$. Higher PS weight $\Rightarrow$ closer PS matches (useful when covariates are high-dimensional and PS captures main confounding).

Matching: design = $c(1, k)$ for 1:k. *exactmatchon* = hard. *finebalanceVars* = soft.

Reg ATE: $\hat{\tau} = \frac{1}{N} \sum [\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i)]$

IPW: $\frac{1}{N} \sum \frac{ZY}{e} - \frac{(1-Z)Y}{1-e}$

AIPW: $\frac{1}{N} \sum [\hat{\mu}_1 - \hat{\mu}_0 + \frac{Z(Y-\hat{\mu}_1)}{e} - \frac{(1-Z)(Y-\hat{\mu}_0)}{1-e}]$

Method	Pro	Con
Match	Transparent	Drops unmatched
Strat	Simple, 90% bias	Stratum noise
IPW	All data, bal any g	Extreme wts
AIPW	Doubly robust	Both wrong=fail